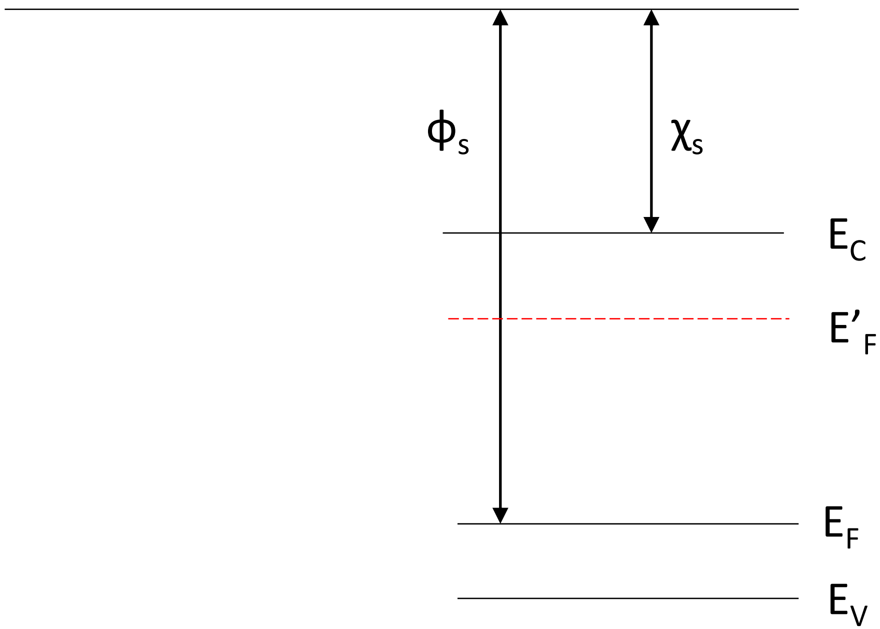
Answer for Slide #13



$$\Phi_s = \chi_s + [\, E_g - (E_F - E_V)] = \chi_s + [\, E_g + kT/q \ln (N_A/N_V)]$$

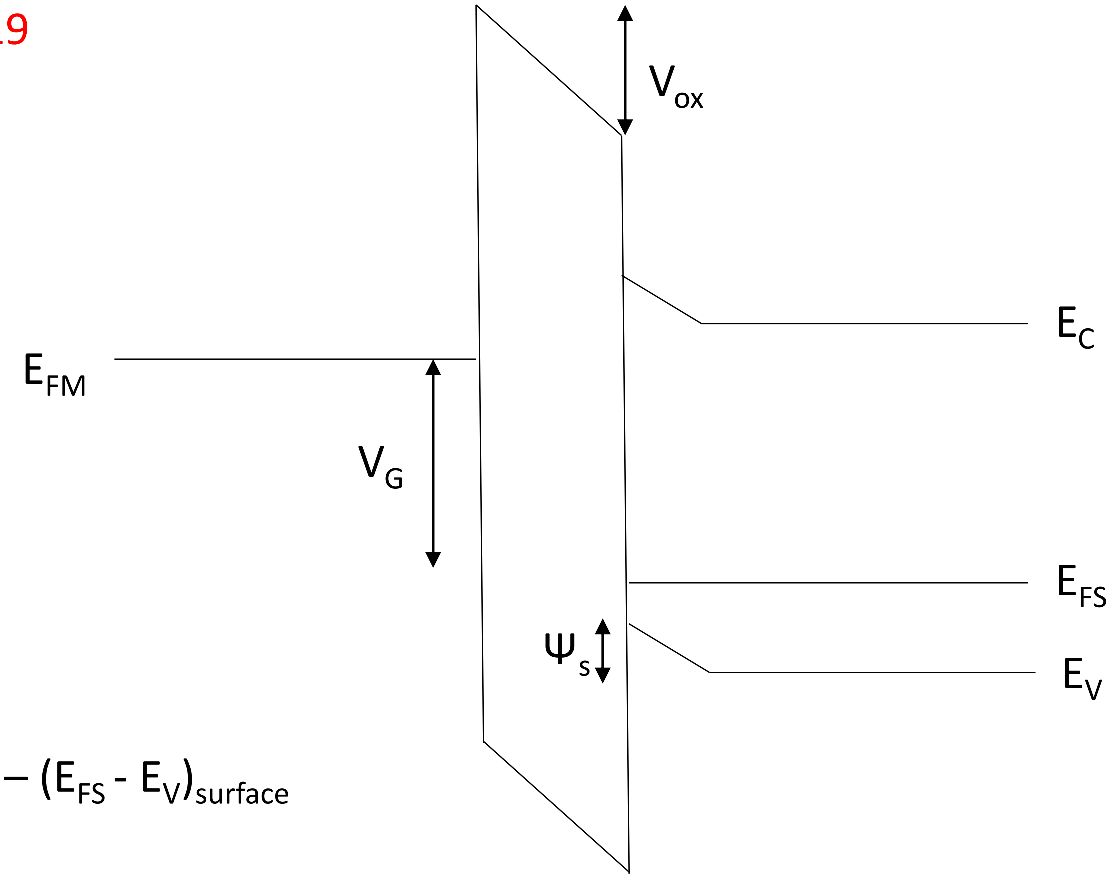
$$= 4.05 + [1.12 + 0.0259 \ln (10^{17}/1.04 \times 10^{19})] = 5.05 \text{ eV}$$

To have the same work function as gate, the work function of Si should be 4.25 eV, which is above the mid bandgap. So, the Si should be n-type.

$$E_C - E'_F = 4.25 - 4.05 = 0.2 \text{ eV}$$

$$N_D = N_C \exp [- (E_C - E'_F)/kT] = 1.24 \times 10^{16} \text{ cm}^{-3}$$

Answer for Slide #19



$V_G = V_{ox} + \Psi_s$

$\Psi_s = (E_{FS} - E_V)_{bulk} - (E_{FS} - E_V)_{surface}$

$(E_{FS} - E_V)_{bulk} = -kT/q \ln (N_A/N_V) = 0.198$ eV

$(E_{FS} - E_V)_{surface} = -kT/q \ln (N_A/N_V) = 0.0786$ eV

$\Psi_s = 0.119$ eV

$V_{ox} = 3 - 0.119 = 2.881$ eV